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LESSON : AWS

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AWS-DEVOPS

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What is NGINX?



NGINX (pronounced “Engine X”) is a free, open-source web server first released in 2004.

It serves websites and can also act as a reverse proxy, load balancer, and HTTP cache.

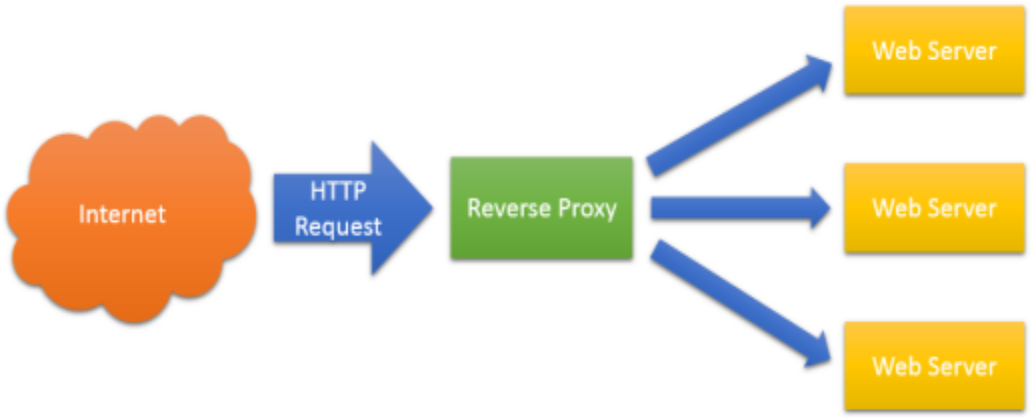
Its event-driven architecture helps manage many connections efficiently.

Official documentation: nginx.org/en/docs/

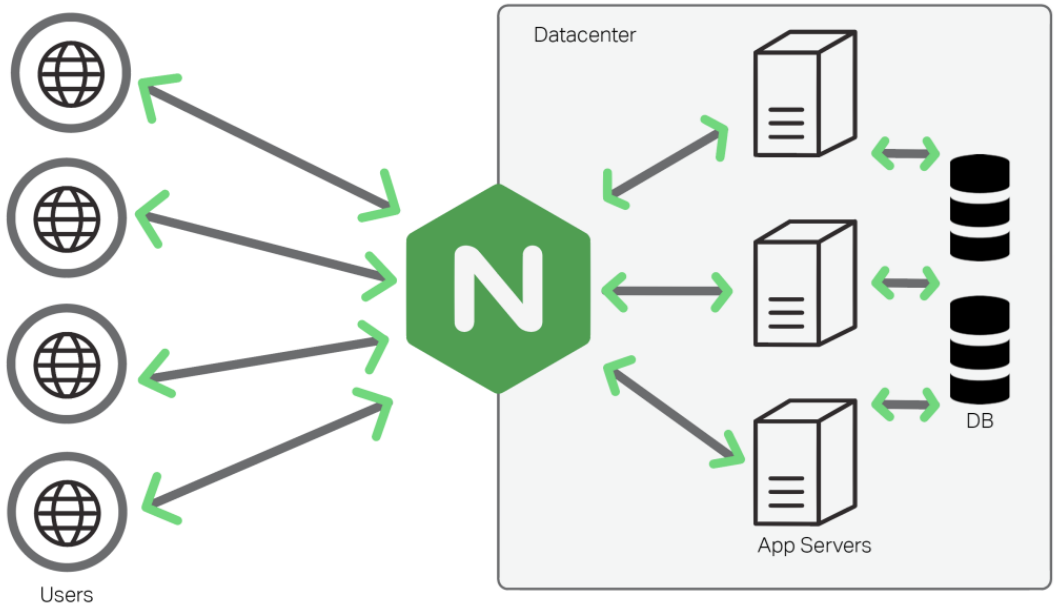
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What is NGINX?

Reverse Proxy



Load Balancing



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What is APACHE HTTP Server?



Apache HTTP Server (httpd) is a free, open-source, cross-platform web server.

The Apache HTTP Server Project began in 1995; it was created by Tim Berners-Lee.

Apache is modular and supports static pages, dynamic applications, reverse proxying, and authentication.

Official documentation: httpd.apache.org/

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Apache vs NGINX



Both NGINX and Apache can serve HTTP requests and act as reverse proxies.

NGINX uses an event-driven worker model; Apache supports multiple processing models (including event).

Neither server is always faster or more secure: results depend on configuration and workload.

Both can be used for small and large websites.

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Apache vs NGINX — Request Handling

NGINX

- Event-driven worker processes
- Efficient connection handling
- Common reverse-proxy use cases
- Handles static content and caching

Apache httpd

- Multiple MPM choices
- event, worker, and prefork modes
- Extensive module ecosystem
- Supports reverse proxy and .htaccess

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Apache vs NGINX — Practical Strengths

NGINX — Common Choices

- Reverse proxy for an application
- Serving images, CSS, and JavaScript
- Routing traffic to several backends
- Caching and connection handling

Apache — Common Choices

- Applications using Apache modules
- Sites needing .htaccess support
- Familiar httpd configuration
- Traditional hosting and PHP stacks

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Apache vs NGINX — Which One Should We Use?

Scenario A: Static Site

A training website contains HTML, CSS, images, and JavaScript.

NGINX or Apache can both serve it successfully.

Try both and compare setup steps.

Scenario B: Node.js App

A Node.js application listens on port 3000.

NGINX or Apache can accept public HTTP requests and reverse-proxy them to the app.

Test reliability and configuration.

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Common NGINX use cases:

- Static websites and assets
- Reverse proxy for Node.js/Python apps
- Load balancing and HTTP caching

Common Apache use cases:

- Modular web hosting
- PHP and application deployments
- Sites using httpd configurations



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NGINX on Amazon Linux 2023

Document root: /usr/share/nginx/html

Main config: /etc/nginx/nginx.conf

Extra configs: /etc/nginx/conf.d/

Apache httpd on Amazon Linux 2023

Document root: /var/www/html

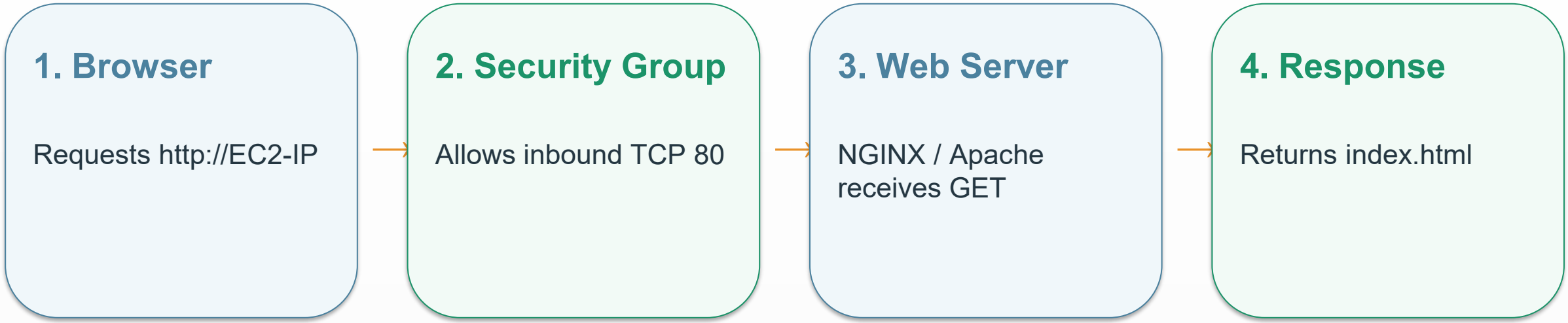
Main config: /etc/httpd/conf/httpd.conf

Extra configs: /etc/httpd/conf.d/



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A Web Request: Browser to EC2



Remember: Port 22 is for SSH administration; port 80 is for HTTP traffic.

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Hands-On Plan — Amazon Linux 2023

NGINX Lab

1. Launch EC2; SSH from My IP
2. Install nginx + git
3. Deploy designer website
4. Test localhost and browser
5. Repeat using EC2 user data

Apache Lab

1. Launch a separate EC2 instance
2. Install httpd + git
3. Deploy same designer site
4. Test localhost and browser
5. Repeat using EC2 user data

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Verification and Troubleshooting

Verify the Server

sudo systemctl status nginx

sudo nginx -t

curl -I http://localhost

Apache alternative:

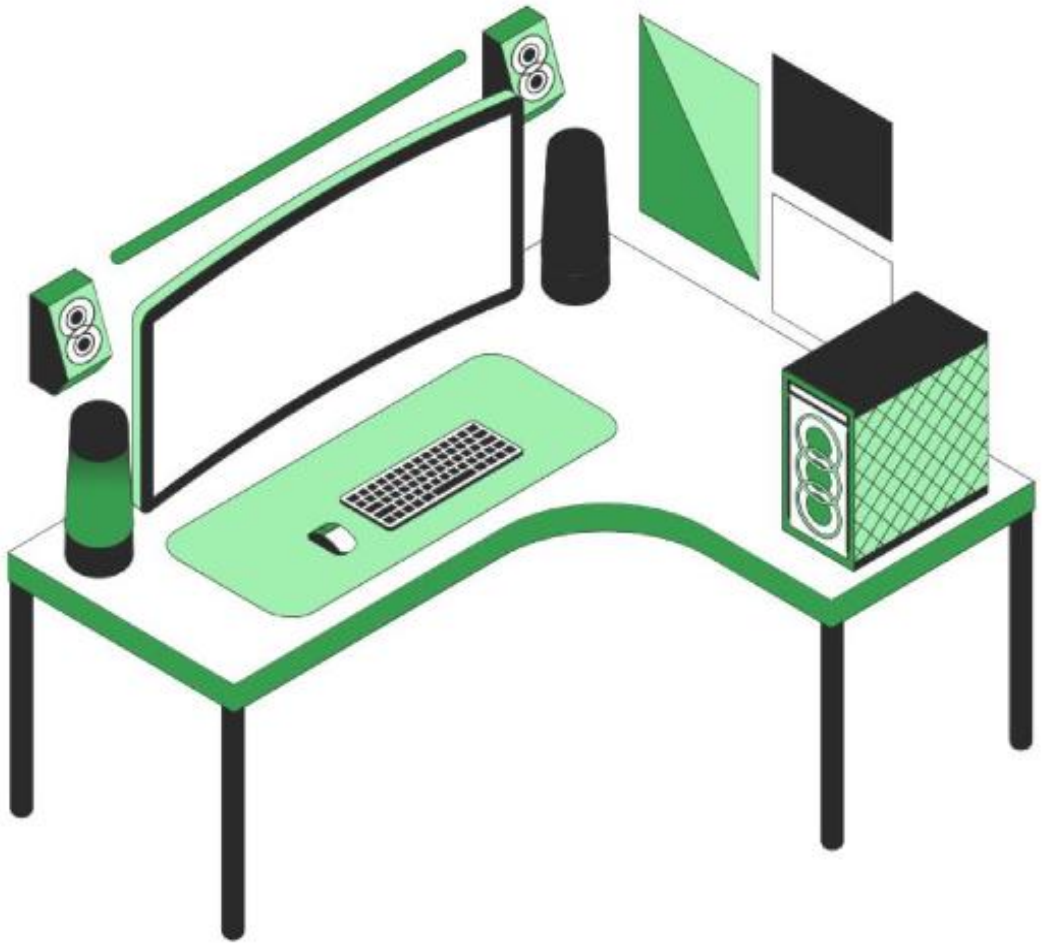
sudo systemctl status httpd

sudo httpd -t

If the Page Does Not Load

- Is port 80 allowed in SG?
- Is the service active?
- Is index.html in the right directory?
- Does EC2 have a public IPv4?
- For user data: check cloud-init logs

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Do you have any questions?

Send it to us! We hope you learned something new.

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